AWT TECHNICAL REGULATIONS 2025

OUR MISSION

TO PROMOTE
REWARDING MANUFACTURING
CAREERS FOR THE FUTURE OF OUR COMMUNITIES

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Revision 2- 10/21/25

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1. GENERAL

1.1. All participants build and operate robots at their own risk. Combat robotics is inherently dangerous. No amount of regulation can encompass all the dangers involved. Please take care not to hurt yourself or others when building, testing, and competing. Robots must be free of words, logos, or graphics that are offensive in any way.

1.2. If you have a robot or weapon design that does not fit within the categories outlined in these rules or is in some way ambiguous or borderline, please contact the Program Coordinator. Safe innovation is always encouraged, but surprising the event staff with your brilliant exploitation of a loophole may cause your robot to be disqualified before it ever competes.

1.3. Compliance with all event rules is mandatory. It is expected that competitors stay within the rules and procedures of their own accord and do not require constant policing.

1.4. Each event has safety inspectors. It is at their sole discretion that your robot is allowed to compete. As a builder, you are obligated to disclose all operating principles and potential dangers to the inspection staff.

2. CARDINAL SAFETY RULES

2.1. Failure to comply with any of the following rules could result in expulsion from the competition. The rules are strict to prevent injury.

2.2. Radios may not be turned on, at or near events, for any purpose without obtaining
permission from the event personnel.

2.3. Proper activation and deactivation of robots is critical. Robots must only be activated in the arena, testing areas, or with the expressed consent of AWT staff or safety officials. Robots and controllers must not be turned on in the pit area; bots can only be tested under the supervision of safety officials.

2.4. All robots must be able to be FULLY deactivated, which includes power to the drive and the weaponry, within 60 seconds by a manual disconnect. (Removable link or Main Power Switch) The operator must be able to perform the deactivation without handling the robot.

2.5. All robots must be able to be fully activated within 30 seconds.

2.6. All robots not in an arena or official testing area must be raised or blocked up in a manner so that their wheels or legs cannot cause movement if the robot were accidentally turned on. Runaway bots are VERY dangerous.

2.7. Locking devices: Moving weapons that can cause damage or injury must have a clearly visible locking pin in place at all times when not in the arena. Locking pins must be painted in neon orange or another high-visibility color. Locking pins must be clearly capable of stopping, arresting, or otherwise preventing the harmful motion of the weapon. When installing/uninstalling locking pins, under no circumstance can a person's body part be in the path of the weapon.

2.8. Weapon locking pins must be in place when the weapon switch is turned on during a robot's power-on procedure (the weapon switch will be on, but no power will be applied to the weapon). The locking pin will be removed just before closing the arena door. This includes all powered weapons regardless of the power source.

2.9. It is expected that all builders will follow basic safety practices during work on the robot at their pit station. Please be alert and aware of your pit neighbors and people passing by. Continued failure to follow safety directions could result in an individual's or the entire team's disqualification from the event.

2.9.1. This includes and is not limited to wearing SAFETY GLASSES at ALL times while in the pit area and in the arena area while handling or controlling your bot. Typical prescription glasses may not be considered safety grade, and it's recommended to wear safety glasses over them.

2.10. Any robots transported outside the pit area must have safety covers/ restraints in place to cover all sharp and or potentially harmful areas on the robot. This will be checked at the robot's safety inspection. Robots can only be activated/ powered on in the test box or arena; everywhere else, the robot must be deactivated.

2.11. SAFETY OFFENSES

2.11.1. Any safety offenses from the safety rules in Section 2 will be handled as follows:

2.11.2. The first safety offense from any member of the team will result in a warning.

2.11.3. The second offense from any member of that same team will result in a 10-second controller impoundment at the beginning of your next match. This means your opponent will be able to attack your immobile robot.

2.11.4. The third offense from any member of the same team will result in a team disqualification from the tournament.

3. WEIGHT CLASSES & SIZE

3.1. This event offers two weight classes: 3lb robots (Junior RoboBots for middle school students) and 15lb robots (RoboBots for high school students).

3.1.1. There is no weight bonus for shufflers or other forms of locomotion that are predicated on rolling - see 4.1.2 for a definition of a non-wheeled robot.

3.2. RoboBots wheeled robot weight = 15lb, JuniorBots wheeled robot = 3lb.

3.3. RoboBots non-wheeled robot weight = 20lb.

3.3.1. If you are considering a non-wheeled robot for JuniorBots, please check with AWT staff.

3.4. Your bot must be no wider than 3 feet and no taller than 3 feet to ensure that your robot fits in the arena door.

3.5. Your bot must be able to clear the arena floor during normal driving operations. Consideration must be made by teams to ensure that their bot will not damage the arena structure.

3.6. Multi-Bots are allowed as long as the combined weight is 15 pounds or less. I

3.6.1. If both bots are the same weight, the team will need to decide which robot is the “primary” robot. If one of the bots weighs more than the other, the heavier bot is automatically the “primary” bot.

3.6.2. The primary bot must be clearly marked with a piece of yellow tape, and the team must inform the referee before each fight which bot is the "primary" bot.

3.6.3. The “primary” bot is identified in case of a countdown or knockout. As long as the “primary” bot is functioning, the match continues. If the “primary” bot is knocked out or it is counted out, the match is over. The other bot cannot compete without the “primary” bot.

4. MOBILITY

4.1. All robots must have easily visible and controlled mobility to compete*. Methods of mobility include:

4.1.1. Rolling (wheels, tracks, or the whole robot)

4.1.2. Non-wheeled: non-wheeled robots have no rolling elements in contact with the floor and no continuous rolling or cam-operated motion in contact with the floor, either directly or via a linkage. Motion is “continuous” if continuous operation of the drive motor(s) produces continuous motion of the robot. Linear-actuated legs and novel non-wheeled drive systems may qualify for this bonus.

4.1.3. Shuffling (rotational cam-operated legs)

4.1.4. Ground effect air cushions (hovercrafts) and flying drones are illegal and forbidden.

4.1.5. In the case of a Multi-bot, only the “primary” bot must have easily visible and controlled mobility to compete.

5. ROBOT CONTROL REQUIREMENTS

5.1. Primary control and fail-safe communications to a bot have to be via a remote radio link. Tethered control is specifically not allowed.

5.1.1. A Bot may be controlled by a maximum of three Operators/Drivers

5.1.2. A bot must have a robust radio fail-safe that shuts off all motion systems and weapons power within one second after the remote-control transmitter is switched off, or otherwise stops transmitting. This fail-safe is required in addition to the Master Switch requirements

5.1.3. Binary (on/off) movement speed control is not allowed. Any control of the bot speed along the ground has to be continuously variable in both forward and reverse directions.

5.2. Spektrum Transmitters are recommended because of their quality and Smart Fail Technology, but there are several other great brands on the market. All transmitters and receivers used in competition MUST have fail-safes. Some additional brands to consider are Radiomasters, FRSky, and Futaba.

5.3. Bluetooth systems must be approved by the safety committee before the competition. Operating plans, schematics, and a clear explanation of controls must be presented for review.

5.3.1. The safety committee should get this information at least one week before the competition so that the safety committee can verify it at the start of the competition.

5.3.2. The bot must comply with all other regulations, meaning the bot should be in a zero-energy state when not in the test box or arena (cage). All power must be off and dissipated. The students should not need to handle the bot to bring the robot to a zero-energy state. The battery must be disengaged by the master switch, and any energy storage devices must automatically drain when the master switch is shut off. The energy storage should only take place when the bot is on. Any capacitors or electrical storage devices used in the system must be capable of being safely discharged without putting the students at risk.

6. AUTONOMOUS /SEMI-AUTONOMOUS ROBOTS

6.1. Any robot that moves, seeks a target, or activates weapons without human control is considered autonomous. If your robot is autonomous, contact the Program Coordinator before the first documentation check-in (January/ Early February).

6.2. Autonomous robots must have a clearly visible light for each autonomous subsystem that indicates whether or not it is in autonomous mode, e.g., if your robot has two autonomous weapons, it should have two “autonomous mode” lights (this is separate from any power or radio indicator lights used).

6.3. The autonomous functionality of a robot must have the capability of being remotely armed and disarmed. (This does not include internal sensors, drive gyros, or closed-loop motor controls.)

6.3.1. While disarmed, all autonomous functions must be disabled.

6.3.2. When activated, the robot must have no autonomous functions enabled, and all autonomous functions must fail-safe to off if there is a loss of power or radio signal.

6.3.3. In case of damage to components that remotely disarm the robot, the robot's autonomous functions are required to automatically disarm within one minute of the match length time after being armed.

7.BATTERIES & POWER

7.1. The only permitted batteries cannot spill or spray any of their contents when damaged or inverted. This means that standard automotive and motorcycle wet cell batteries are prohibited.

- 7.1.1. Examples of batteries that are permitted: gel cells, Hawkers, NiCads, NiMh, dry cells, LiFePO4, AGM, and LiIon.
- 7.1.2. NO LiPoly batteries will be allowed.
- 7.1.3. If your design uses a new type of battery, or one you are not sure about, please contact the Program Coordinator.

7.2. TBD

7.3. All electrical power to any sub-system (weapon and drive systems that could cause potential human bodily injury) must be able to fully deactivate without endangering the person turning off the robot/ system(E.g, no body parts in the way of weapons or pinch points) within 15 seconds.

- 7.3.1. The shutdown or deactivation must be a manually operated mechanical method, such as a power switch like Whyachi, Hella, other mechanical switches, or a removable link.
- 7.3.2. Relays may be used to control the power, but there must also be a mechanical disconnect method.
- 7.3.3. Please refer to section 2.4 for the complete shutdown time.

7.4. All efforts must be made to protect battery terminals from a direct short, thus causing a battery fire. All charging of batteries must be done outside of the bot. It is also recommended to charge all batteries in a fireproof battery bag. All robots must have a separate light per switch that is easily visible from the outside of the robot, showing that its main power and or sub-system is activated.

7.5. All robots must have a separate light per switch that is easily visible from the outside of the robot, showing that its main power and or sub-system is activated.

7.6. Batteries must be visible for inspection and must have a marking from the manufacturer that clearly identifies the type of battery. If such markings are not possible, be prepared to show another form of proof that your battery is allowed (vendor receipt, etc).

7.7. Primary-power batteries and any power-handling electrolytic-type capacitors must be securely mounted and enclosed within the structural frame of the robot.

- 7.7.1. **Zip ties are allowed** for mounting batteries.
- 7.7.2. **Duct tape and Velcro are not** approved methods for securing batteries.
- 7.7.3. If you're considering another method, please contact AWT staff and the safety team.

8.PNEUMATICS

8.1. Robots in both the RoboBots and Junior RoboBots classes are NOT allowed to use pneumatics.

9. HYDRAULICS

9.1. Robots in both the RoboBots and Junior RoboBots classes are NOT allowed to use hydraulics.

10. INTERNAL COMBUSTION ENGINES (ICE) & LIQUID FUELS

10.1. Robots in both the RoboBots and Junior RoboBots classes are NOT allowed to use ICE.

11. ROTATIONAL WEAPONS/ FULL-BODY SPINNING ROBOTS

11.1. Spinning weapons cannot contact the outer arena walls during normal operation. (Contact with an inner arena curb, or containment wall, is allowed).

11.2. Spinning weapons must come to a full stop within 60 seconds of the power being removed.

12. SPRINGS & FLYWHEELS

12.1. Springs used in robots will use the remaining rules in this section. Safe operation, good engineering, and best practices must be used in all systems.

12.2. Any springs used for drive or weapon power must have a way of loading and actuating the spring remotely under the robot's power.

12.3. Springs used for active weapons must not be loaded when the robot is out of the arena or testing area.

12.4. Springs used within switches or other internal operations are exempt from this rule.

12.5. Any flywheel or similar kinetic energy storing device must not be spinning or storing energy in any way unless inside the arena or testing area.

12.5.1. There must be a way of generating and dissipating the energy from the device remotely under the robot's power.

12.6. All springs, flywheels, and similar kinetic energy-storing devices must fail to a safe position on loss of radio contact or power.

13. FORBIDDEN WEAPONS & MATERIALS

13.1. The following weapons and materials are absolutely forbidden from use

13.1.1. Weapons designed to cause invisible damage to the other robot. This includes but is not limited to:

13.1.1.1. Electrical weapons

13.1.1.2. RF jamming equipment, etc.

13.1.1.3. EMF fields from permanent or electromagnets that affect another robot's electronics.

13.1.1.4. Weapons or defenses that stop the combat completely of both (or more) robots. This includes nets, tapes, strings, and other entanglement devices.

13.2. Weapons that require significant cleanup or, in some way, damage the arena require repair for further matches. This includes but is not limited to:

13.2.1. Liquid weapons. Additionally, a bot may not have liquid that can spill out when the robot is superficially damaged.

13.2.2. Foams and liquefied gases.

13.2.3. Powders, sand, ball bearings, and other dry chaff weapons

13.3. Un-tethered Projectiles (see tethered projectile description in Special Weapons section 14.1)

13.4. Heat and fire are forbidden as weapons. This includes, but is not limited to, the following:

13.4.1. Heat or fire weapons are not specifically allowed in the Special Weapons section

13.4.2. Flammable liquids or gases

13.5. Explosives or flammable solids such as:

13.5.1. DOT Class C devices

13.5.2. Gunpowder / Cartridge Primers

13.5.3. Military Explosives, etc.

13.6. Light and smoke-based weapons that impair the viewing of robots by an Entrant, Judge, Official, or Viewer. (You are allowed to physically engulf your opponent with your robot, however.) This includes, but is not limited to, the following:

13.6.1. Smoke weapons not specifically allowed in the Special Weapons section.

13.6.2. Lights such as external lasers above 'class I' and bright strobe lights.

13.7. Hazardous materials are forbidden from use anywhere on a robot where they may make contact with electrical components or batteries.

14. SPECIAL WEAPON DESCRIPTION ALLOWED

14.1. Tethered Projectiles are allowed, but must be no longer than 3 feet and may not entangle the opponent.